

PRACTICAL NO:7

AIM: Sampling and Sampling Distributions:

- Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.
- Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

(a) Conduct a Survey and Collect Sample Data

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sample(a).py - C:/Users/info/OneDrive/Desktop/prac 6 decision/sample(a).py (3.14.6)
File Edit Format Run Options Window Help
print("T110 SANJANA SAHU")
#(a) Conduct a Survey and Collect Sample Data
import numpy as np
import matplotlib.pyplot as plt

# Sample data
data = [2, 3, 4, 2, 5, 3, 4, 6, 3, 2,
        4, 5, 3, 4, 2, 6, 5, 3, 4, 2,
        3, 5, 4, 6, 3, 2, 4, 5, 3, 4]

# Calculate sample mean
sample_mean = np.mean(data)

print("Sample Size:", len(data))
print("Sample Mean:", sample_mean)

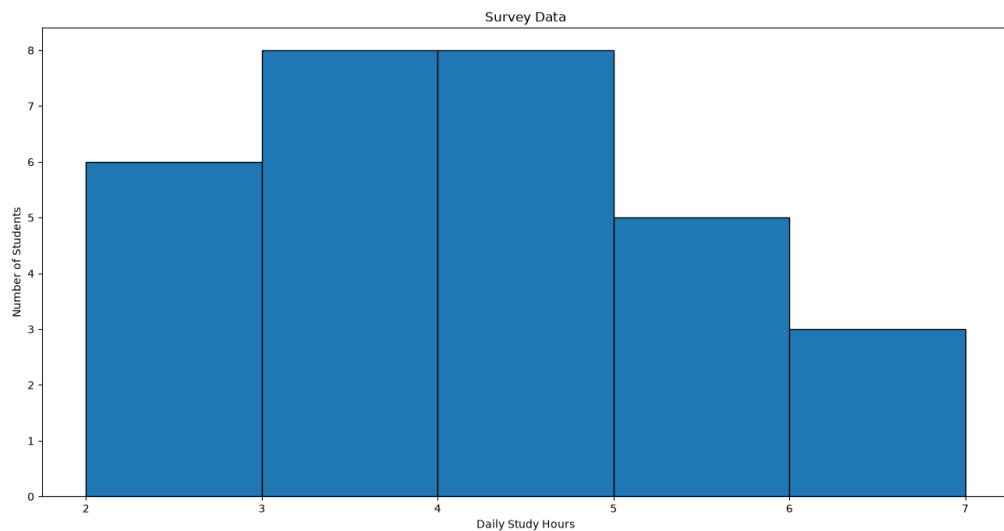
plt.hist(data, bins=range(2, 8), edgecolor='black')
plt.xlabel("Daily Study Hours")
plt.ylabel("Number of Students")
plt.title("Survey Data")
plt.show()

Ln: 2 Col: 45

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v
.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/info/OneDrive/Desktop/prac 6 decision/samp
le(a).py =====
Sample Size: 30
Sample Mean: 3.7

Ln: 5 Col: 0
```



(b) Central Limit Theorem

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```
FILE SAMPLE(B).py - C:/Users/info/OneDrive/Desktop/prac 6 decision/SAMPLE(B).py (3.14.6)
File Edit Format Run Options Window Help
print("T110,SANJANA SAHU")
#(b) Central Limit Theorem
import numpy as np
import matplotlib.pyplot as plt

# Population
population = np.array([
    2, 3, 4, 2, 5, 3, 4, 6, 3, 2,
    4, 5, 3, 4, 2, 6, 5, 3, 4, 2,
    3, 5, 4, 6, 3, 2, 4, 5, 3, 4
])

# Generate sample means
sample_means = []

for i in range(1000):
    sample = np.random.choice(population, size=10, replace=True)
    sample_means.append(np.mean(sample))

# Mean of sampling distribution
mean_of_means = np.mean(sample_means)
std_of_means = np.std(sample_means)

print("Mean of Sampling Distribution:", mean_of_means)
print("Standard Deviation of Sampling Distribution:", std_of_means)

# Plot
plt.hist(sample_means, bins=30, edgecolor='black')
plt.xlabel("Sample Means")
plt.ylabel("Frequency")
plt.title("Sampling Distribution of Sample Mean")
plt.show()
```

```
FILE IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 1
0:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more i
nformation.
>>>
===== RESTART: C:/Users/info/OneDrive/Desktop/prac
6 decision/SAMPLE(B).py =====
T110,SANJANA SAHU
Mean of Sampling Distribution: 3.6927
Standard Deviation of Sampling Distribution: 0.3824
8752920846973
|
```

Figure 1

